

NCERT

National Cyber Emergency Response Team

DIGITAL FORENSICS LABORATORY — GROUP PROJECT REPORT

Evidence Management System (with Chain of Custody)

Phase 3 Report — Final Modules, Verification, and Project Completion
100% Project Completion

Group:	Group 2
Members:	Muhammad Ozair Khalid
	Zakia Qayyum
	Uniba Fatima
Instructor:	Sir Umar Javaid
Course:	Digital Forensics Internship — Project Completion Phase 3
Platform:	Next.js 14 · PostgreSQL · Prisma · NextAuth · SHA-256 · PDF Reports
Submission:	July 2026
Progress:	100% Complete (Submitted Deliverable)

Academic Integrity Notice: All design, implementation, and verification activities were performed on an isolated, instructor-authorized laboratory / development system for the NCERT Digital Forensics Internship. No production case systems or third-party operational evidence stores were accessed at any stage of this project.

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Abstract

This report documents **Phase 3 — final completion (100%)** of Group 2's NCERT Digital Forensics Internship project *Evidence Management System (with Chain of Custody)*, supervised by **Sir Umar Javaid**. Phase 3 delivers the remaining half of the platform: handoff re-hashing with integrity flags and supervisor resolution, printable custody PDF reports, a full filterable audit trail with export, admin user management, and presentation-ready landing/login surfaces. Automated smoke verification, TypeScript checks, and role-permission tests confirm that all four bullets of the official Project 1 brief are satisfied. The system is ready for internship demonstration and submission.

1 Introduction

Phases 1 and 2 produced research, architecture, authentication, evidence ingest with SHA-256, and a custody ledger. Phase 3 closes the forensic loop: **prove integrity on movement, package custody history for printing, retain a system-wide audit trail, and administer users under least privilege**. Completion is claimed only after verification against the Project 1 brief.

1.1 Objectives

- 1 Implement re-hash on custody handoff with mismatch flagging and movement blocks.
- 2 Provide supervisor resolution of integrity flags with written notes.
- 3 Generate printable per-item custody reports (PDF) including hashes.
- 4 Expose a full audit trail UI with filters and CSV export.
- 5 Deliver admin user create / role change / soft-deactivate flows.
- 6 Polish public landing and login for NCERT demonstration quality.
- 7 Run smoke, typecheck, build, and role-matrix verification; map results to the brief.

1.2 Scope

Phase 3 covers all remaining functional brief items and final verification. Hardware acquisition, mobile device imaging, and national case-management integration remain outside the internship deliverable, consistent with the original “digital evidence inventory + custody ledger” framing.

1.3 Tools and Environment

Component	Details
Application	Next.js 14 + TypeScript + Tailwind + shadcn/ui
Data	PostgreSQL + Prisma (migrate, seed)
Auth / RBAC	NextAuth JWT + can()/requirePermission()
Integrity	Server-side SHA-256 recompute on handoff
Reports	@react-pdf/renderer custody package
Verification	npm run smoke, tsc --noEmit, lint, build
Demo URL	http://localhost:3000 (lab environment)

Table 1: Final toolchain for Phase 3 completion and verification.

2 Continuity from Phases 1–2

Open item after Phase 2	Phase 3 delivery
FR-06 Re-hash on handoff / flag mismatches	Integrity module + custody gate
FR-07 Supervisor resolution	/integrity resolve workflow
FR-08 Printable custody report	PDF generate + download
FR-10 Full audit trail	/audit table, filters, CSV export
Admin user lifecycle	/admin/users
Demo surface polish	Landing page + upgraded login

Table 2: How Phase 3 closes open requirements from earlier phases.

3 Module Implementation — Remaining 50%

3.1 Integrity Re-Hash and Flag Resolution

On custody handoffs involving file-backed exhibits, the server recomputes SHA-256 and compares it to the baseline digest. A mismatch sets status `INTEGRITY_FLAGGED`, records the check, surfaces a hard warning, and blocks routine further transfers until a supervisor (or admin) records a written resolution. This implements the brief’s “re-hash on each handoff to flag data mismatches” requirement without ever silently replacing the original hash.

3.2 Printable Custody Reports (PDF)

From the reports workflow (and evidence detail), authorised roles generate a custody report PDF containing exhibit identity, baseline hash (monospace), chronological custody events, and integrity outcomes. The layout follows a formal laboratory style suitable for packaging with case materials during internship demonstration.

3.3 Full Audit Trail

Security-relevant actions—login, evidence create, custody events, integrity flags/resolutions, report generation, user administration—append to `AuditLogEntry`. The `/audit` module provides filtering, expandable metadata, and CSV export. There is no update/delete API for audit rows, preserving append-only semantics argued in Phase 1.

3.4 Admin User Management

Administrators manage the user directory: create accounts (temporary password shown once), change roles, and soft-deactivate users. This supports sustainable lab use beyond the four seeded personas and completes organisational control around the three brief-named roles plus admin.

3.5 Landing, Login Polish, and UX Hardening

A public landing page explains the platform’s forensic value (hashing, custody, RBAC, audit) for NCERT audiences. The login experience includes demo account affordances for rapid role switching during viva-style walkthroughs. Loading skeletons, empty states, and module accents improve clarity without changing security behaviour.

4 End-to-End Verification

#	Verification	Result
1	npm run smoke (routes, RBAC matrix, PDF render, actions)	PASSED
2	npx tsc --noEmit	PASSED
3	npm run lint	PASSED
4	npm run build	PASSED
5	Register → transfer → mismatch → resolve → PDF → audit	PASSED (manual)
6	Custodian denied /admin and /audit	PASSED
7	Public / and /login HTTP 200; protected routes gated	PASSED

Table 3: Automated and manual verification results at completion.

5 Brief Compliance Matrix (100%)

Brief requirement	System fulfilment	Primary modules
Register items with unique IDs, intake, SHA-256 at ingest	Complete	/evidence, /evidence/new
Record seizure/transfer/examination/return with handler, time, location, reason	Complete	/custody + evidence timeline
Re-hash on handoff; flag mismatches; printable custody report	Complete	/integrity + PDF reports
RBAC (custodian, examiner, supervisor) + full audit trail	Complete (+ admin)	middleware, /audit, /admin/users

Table 4: Official Project 1 scope versus final system (complete).

Phase	Weight	Outcome
Phase 1	25%	Research, requirements, architecture, RBAC/data design
Phase 2	25% → 50%	Auth, evidence+SHA-256, custody ledger, dashboard core
Phase 3	50% → 100%	Integrity, PDF, audit, admin, polish, verification
Project status	100%	Ready for NCERT internship submission / demo

Table 5: Cumulative phase progress to 100%.

6 Results Summary

The completed system is a working Evidence / inventory management application: exhibits receive unique IDs and SHA-256 baselines; every custody event is attributed and timed; handoffs re-hash and flag mismatches; supervisors resolve flags; printable reports package the ledger; and roles plus audit logging protect and explain system activity. Automated smoke verification and manual scenario testing support the 100% completion claim.

7 Conclusion

Across three phases, Group 2 researched, built, and verified an Evidence Management System with Chain of Custody aligned to the NCERT Project 1 brief. Phase 1 justified the forensic controls; Phase 2 implemented the operational spine; Phase 3 closed integrity, reporting, audit, and administration. Under

the supervision of Sir Umar Javaid, the team submits this Phase-3 report as the formal completion record of the internship software deliverable.

References

- [1] Group 2 — Phase 1 Report: Research & Foundation (25%), NCERT Digital Forensics Internship.
- [2] Group 2 — Phase 2 Report: Core Modules (50%), NCERT Digital Forensics Internship.
- [3] NCERT Project 1 brief: Evidence Management System (with Chain of Custody).
- [4] NIST FIPS 180-4 — Secure Hash Standard.
- [5] E. U. Haque et al., “Cyber forensic investigation infrastructure of Pakistan,” *IEEE Access*, 2023.
- [6] Next.js / NextAuth / Prisma / @react-pdf/renderer official documentation.
- [7] SWGDE Best Practices for Computer Forensics.
- [8] Group 2 Week 3 NCERT laboratory report — Digital Forensic Investigation Report format reference.

A Smoke / Demo Checklist

Step	Action	Expected result
1	Login as examiner1@ems.local	Dashboard accessible
2	Register evidence with file upload	EVD-ID + SHA-256 stored
3	Log TRANSFER to custodian	Timeline + toast confirmation
4	Alter bytes; attempt handoff	Integrity flag; movement blocked
5	Login as supervisor; resolve flag	Written resolution; audit entry
6	Generate custody PDF	Downloadable report with hashes
7	Open /audit as supervisor	Ordered activity including mismatch
8	Login as custodian; hit /admin	Denied / redirected

Table 6: Recommended instructor demonstration sequence.

B Role Permission Matrix (Final)

Area	Custodian	Examiner	Supervisor	Admin
Dashboard / Evidence / Custody	Yes	Yes	Yes	Yes
Integrity re-hash	No	Yes	Yes	Yes
Integrity resolve	No	No	Yes	Yes
Reports (PDF)	No	Yes	Yes	Yes
Audit trail	No	No	Yes	Yes
Admin users	No	No	No	Yes
Account settings	Yes	Yes	Yes	Yes

Table 7: Final RBAC matrix verified by smoke tests.